Phase Changes and Heating Curves

Answer key with worked solutions

A. Ideas

1. Phase changes.

> (a) **condensation**, exothermic (b) **sublimation**, endothermic (c) **evaporation** (vaporization), endothermic (d) **deposition**, exothermic (e) **freezing**, exothermic

2. Constant temperature.

> The heat supplied is used to **overcome the intermolecular forces** (hydrogen bonds) as liquid turns into vapour, not to make the molecules move faster. The temperature only rises once all the water has boiled away, or if the pressure changes.

B. One-step calculations

3. Melting and freezing.

> (a) $n = \dfrac{100.\ \text{g}}{18.02\ \text{g/mol}} = 5.549\ \text{mol}$; $q = 5.549\ \text{mol} \times 6.01\ \text{kJ/mol} = \mathbf{33.4\ kJ}$
> (b) $n = \dfrac{50.0\ \text{g}}{18.02\ \text{g/mol}} = 2.775\ \text{mol}$; $q = 2.775\ \text{mol} \times 6.01\ \text{kJ/mol} = 16.7\ \text{kJ}$, **released** (q = −16.7 kJ for the water).

4. Ethanol.

> $n = \dfrac{10.0\ \text{g}}{46.07\ \text{g/mol}} = 0.2171\ \text{mol}$; $q = 0.2171\ \text{mol} \times 38.6\ \text{kJ/mol} = \mathbf{8.38\ kJ}$

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C. Multi-step calculations

5. Ice at −20 °C to water at 50 °C.

> Warm ice: $q_1 = 10.0\ \text{g} \times 2.09\ \text{J/(g·°C)} \times 20.0\ \text{°C} = 418\ \text{J} = 0.418\ \text{kJ}$
>
> Melt: $q_2 = \dfrac{10.0\ \text{g}}{18.02\ \text{g/mol}} \times 6.01\ \text{kJ/mol} = 3.335\ \text{kJ}$
>
> Warm water: $q_3 = 10.0\ \text{g} \times 4.184\ \text{J/(g·°C)} \times 50.0\ \text{°C} = 2092\ \text{J} = 2.092\ \text{kJ}$
>
> Total: $(0.418 + 3.335 + 2.092)\ \text{kJ} = \mathbf{5.85\ kJ}$

6. Ice in a drink.

> Melt: $\dfrac{200.\ \text{g}}{18.02\ \text{g/mol}} \times 6.01\ \text{kJ/mol} = 66.70\ \text{kJ}$
>
> Warm: $200.\ \text{g} \times 4.184\ \text{J/(g·°C)} \times 20.0\ \text{°C} = 16\,736\ \text{J} = 16.74\ \text{kJ}$
>
> Total: $66.70\ \text{kJ} + 16.74\ \text{kJ} = \mathbf{83.4\ kJ}$. Most of the cooling comes from melting.

D. Vapour pressure and boiling

7. Everest.

> Water boils when its **vapour pressure equals the external pressure**. With the air pressure only about one third of an atmosphere, the vapour pressure of water reaches it at a much lower temperature (about 70 °C). The water cannot get hotter than its boiling point, so food cooks more slowly.

E. Challenge

8. Steam versus water.

> Water, 100 °C → 37 °C: $10.0\ \text{g} \times 4.184\ \text{J/(g·°C)} \times 63\ \text{°C} = 2636\ \text{J} = \mathbf{2.64\ kJ}$
>
> Steam: condensing $\dfrac{10.0\ \text{g}}{18.02\ \text{g/mol}} \times 40.7\ \text{kJ/mol} = 22.59\ \text{kJ}$, then cooling 2.64 kJ: total **25.2 kJ**.
>
> The steam releases almost **ten times** more heat, because condensing releases the large enthalpy of vaporization directly onto the skin.

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